

2. API Endpoint Plan

The RaceDay API provides endpoints for authentication, user profiles, events, event categories, participant enrolments, and race results. Each endpoint specifies the HTTP method, route, purpose, required user role, request body, and expected response codes.

2.1 Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as an Organiser or Participant.	Public	fullName, email, password, phoneNumber, role	201 Created – User registered successfully. 400 Bad Request – Invalid data or email already exists.
POST	/api/auth/login	Authenticates a user and provides an access token.	Public	email, password	200 OK – JWT token and user role returned. 401 Unauthorized – Invalid credentials.

2.2 User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Retrieves the profile of the currently logged-in user.	Authenticated User	None	200 OK – User profile returned. 401 Unauthorized – User is not logged in.
PUT	/api/users/profile	Updates the logged-in user's profile details.	Authenticated User	fullName, phoneNumber	200 OK – Profile updated. 400 Bad Request – Invalid data. 401 Unauthorized – User is not logged in.

2.3 Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves a list of all upcoming events.	Public	None	200 OK – Array of events returned.
GET	/api/events/{id}	Retrieves details of a specific event.	Public	None	200 OK – Event returned. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new race event.	Organiser	eventName, eventDate, location, description	201 Created – Event created. 400 Bad Request – Invalid event data. 401 Unauthorized – User is not authenticated.
PUT	/api/events/{id}	Updates an existing event owned by the organiser.	Organiser (Owner)	eventName, eventDate, location, description	200 OK – Event updated. 403 Forbidden – User is not the event owner. 404 Not Found – Event does not exist.
DELETE	/api/events/{id}	Deletes an existing event owned by the organiser.	Organiser (Owner)	None	200 OK – Event deleted. 403 Forbidden – User is not the event owner. 404 Not Found – Event does not exist.

2.4 Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Retrieves all race categories belonging to an event.	Public	None	200 OK – Array of categories returned. 404 Not Found – Event does not exist.
POST	/api/events/{eventId}/categories	Adds a new race category to an event.	Organiser (Owner)	categoryName, distanceKm, entryFee, maxParticipants	201 Created – Category created. 403 Forbidden –

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					User is not the event owner. 404 Not Found – Event does not exist. 400 Bad Request – Invalid category data.
PUT	/api/categories/{id}	Updates an existing race category.	Organiser (Owner)	categoryName, distanceKm, entryFee, maxParticipants	200 OK – Category updated. 403 Forbidden – User is not the event owner. 404 Not Found – Category does not exist.
DELETE	/api/categories/{id}	Removes a race category from an event.	Organiser (Owner)	None	200 OK – Category deleted. 403 Forbidden – User is not the event owner. 404 Not Found – Category does not exist.

2.5 Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories/{categoryId}/enrol	Enrols the logged-in	Participant	None – Participant	201 Created – Enrolment

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
		participant into a race category.		is identified from the JWT token.	created. 404 Not Found – Category does not exist. 409 Conflict – Participant is already enrolled.
GET	/api/enrolments/my	Retrieves all race enrolments belonging to the logged-in participant.	Participant	None	200 OK – Array of enrolments returned. 401 Unauthorized – User is not authenticated.
GET	/api/events/{eventId}/enrolments	Retrieves all participant enrolments for an event.	Organiser (Owner)	None	200 OK – Array of enrolments returned. 403 Forbidden – User is not the event owner. 404 Not Found – Event does not exist.
DELETE	/api/enrolments/{id}	Cancels an enrolment belonging to the logged-in participant.	Participant (Owner)	None	200 OK – Enrolment cancelled. 403 Forbidden – Enrolment does not belong to the user. 404 Not Found – Enrolment does not exist.

2.6 Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/results	Records a race result for a participant's enrolment.	Organiser (Owner)	finishTime, position	201 Created – Result recorded. 403 Forbidden – User is not the event owner. 404 Not Found – Enrolment does not exist. 409 Conflict – Result already exists.
PUT	/api/results/{id}	Updates an existing race result.	Organiser (Owner)	finishTime, position	200 OK – Result updated. 403 Forbidden – User is not the event owner. 404 Not Found – Result does not exist.
GET	/api/results/my	Retrieves the logged-in participant's personal race results.	Participant	None	200 OK – Array of results returned. 401 Unauthorized – User is not authenticated.
GET	/api/events/{eventId}/results	Retrieves all race results for an event, such as for a public leaderboard.	Public	None	200 OK – Array of results returned. 404 Not Found – Event does not exist.

